

Nets vs. Sequences

Sequence: Indexed by $\mathbb{N}$

$$(x_n)_{n \in \mathbb{N}} = (x_0, x_1, x_2, \dots)$$

Limited: sequences insufficient for general topologies



Generalization

Net: Indexed by a Directed Set

$(x_a)_{a \in A}$ where A is a directed set

General: applies to all topological spaces